

A Study on the Evolutionary Mechanism of Human High Intelligence

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Abstract

Humanity's rise to the apex of the ecosystem is conventionally attributed to superior physical traits, early tool use, or random evolutionary luck.

However, these explanations fail to answer why other primates in identical ecological niches did not evolve similar cognitive capacities. This paper presents a novel mathematical and biomechanical framework proving that human high intelligence was not an innate superiority, but a desperate paradox forced by extreme physical vulnerability. By modeling the

Collective Imitation Survival Formula and the **Biomechanical Energy Efficiency Formula**, we demonstrate that physical weakness forced early hominins onto the savanna, driving an explosive reliance on collective information processing. The resulting brain expansion induced severe mechanical strain on a quadrupedal spine, compelling the evolution of bipedalism as an energy-conserving necessity ($\theta \rightarrow 89^\circ$). Thus, physical weakness was the prerequisite that unlocked human cognitive evolution, laying the hardware foundation for subsequent tools and fire control.

1. Introduction

Humans entered the harsh survival arena of the ecosystem empty-handed, lacking any inherent biological weaponry. Yet, humanity learned to observe the environment, fashion tools, and ultimately master fire—the decisive catalyst that maximized survival probability. Powered by fire, humans built

the towers of civilization and evolution, finally reaching the apex of the ecological pyramid. Today, standing uncontested at the summit, humans look down from the top, creating advanced technologies not merely for survival, but for convenience and comfort.

When asking, "*What was the primary driver that propelled humanity to the top of the ecosystem?*" the public often responds that humans were blessed with a sacred privilege or possessed bodies naturally optimized for tool use. Meanwhile, traditional evolutionary anthropology and archaeology define the rise of *Homo sapiens* through explosive cognitive development and tool use, yet offer vague, macroeconomic explanations such as sudden climate shifts or accidental adaptive stimuli.

However, these conventional perspectives fail to answer a fundamental question: **"Why did other primates (such as chimpanzees and gorillas) sharing similar African savanna environments fail to evolve into high-intelligence species?"** Furthermore, regarding the sequence of causality prior to full cognitive awakening, they offer only unclear conjectures as to **"Why humans invented and utilized tools in the first place."**

Accordingly, this paper proves that human high intelligence was by no means a product of chance. Instead, we mathematically demonstrate that it was the result of desperate survival calculations and adaptation undertaken by early human ancestors—the physically weakest among contemporary primates—to survive within high-density predation pressure.

2. Main Body

2.1 Ecological Pressure in the Ancient African Savanna and the Formation of the Collective Imitation Network

The ancient African savanna where *Homo sapiens* first emerged was a

high-density competitive zone populated by far more diverse and powerful predator populations than today. In this fragile environment, the physical conditions of human ancestors were utterly powerless. They lacked camouflage or ambush capabilities like small animals (e.g., frogs, snails), lacked the sprinting speed to escape predators like medium-sized herbivores (e.g., gazelles), and lacked the physical strength to overpower apex predators like lions or crocodiles. From a predator's perspective, early humans were optimal, highly efficient prey—offering substantial energy yield without defensive capabilities, compared to fast targets that required high capture energy for minimal nutrition.

The absolute objective of an organism is individual survival and the preservation of genetic information. Most primates chose to maximize their survival rate in trees to evade terrestrial predation, developing extreme upper-body musculature and grip strength. However, the direct ancestral line of humans lacked the metabolic energy required for such physical adaptations and possessed the weakest physical capabilities among the primate family. As a result, they lost the competition for arboreal territory to other primates and were forced down to the ground, facing an evolutionary dead end exposed defenselessly to terrestrial predators.

Desperately seeking a chance at survival, human ancestors abandoned individual physical enhancement and instead adopted an unprecedented social survival strategy: maximizing cooperation and the **Collective Imitation Network**. This mathematical survival model is formulated in **[Formula 1: Collective Imitation Survival Rate Formula]**:

$$S = K_0 + \frac{(1 - K_0) \times (E_v \times M \times N)}{D}$$

S : Final Survival Rate

K_0 : Individual Pure Physical Capability & Defense (Base Kinematics)

D : Environmental Danger / Predation Pressure

E_v : Environmental Adaptation Vector

M : Imitation & Learning Efficiency

N : Group Population / Network Size

This formula clearly explains the human evolutionary mechanism. Due to the characteristics of the African savanna, danger (D) was extremely high, while human initial physical capability (K_0) converged near zero. With individual survival at catastrophic levels, the only mathematical way for humans to raise their overall survival rate (S) was to exponentially expand the numerator—the collective imitation network term representing physical deficit compensation: $(1 - K_0) \times M \times N$.

As observed in feral child cases, humans combined their unique imitation engine (M)—which rapidly absorbs the survival skills of other individuals—with group size (N) to exponentially spread and accumulate fragmented survival knowledge. In processing this massive influx of information, the human brain inevitably took on immense metabolic burdens and computational workloads.

Mathematical Verification and Simulation of Formula 1 using Ecological Data

In this formula, the environmental vector (E_v) is calculated based on the savanna flatlands (grasslands) rather than the canopy. This accounts for the ground-level hazard zone where human ancestors were isolated after losing arboreal competition, proving that this severe penalty acted as the paradoxical trigger activating collective imitation. Substituting academic

baseline data under savanna danger ($D = 10$) and ground adaptation ($E_v = 0.2$) yields:

Apex Predators (Lion Pride):

$$K_0 = 0.95, M = 0, N = 5 \Rightarrow S = 0.95$$

Interpretation: Lions maintain high survival (S) purely through overwhelming physical dominance (K_0). Thus, they lack any evolutionary incentive to operate a collective imitation network for knowledge transmission.

Early Hominins (Homo Lineage Group):

$$K_0 = 0.1, M = 0.9, N = 25 \Rightarrow S = 0.505$$

Interpretation: With the worst physical defense on the open savanna, humans activated their imitation efficiency (M) and group network size (N) to extreme levels—the sole mathematical key to pulling their survival rate into safe territory.

2.2 Biomechanical Limits of Brain Expansion and Structural Reconstruction via Bipedalism

As limited metabolic energy concentrated into the brain due to social computation and knowledge accumulation, hominin brain volume inevitably expanded. Biologically, however, increased brain mass and volume induced catastrophic mechanical overloads on the cervical and spinal vertebrae of quadrupedal apes. Supporting a heavier head during terrestrial locomotion created a fatal energy leak.

Unable to tolerate this metabolic waste, human ancestors undertook a radical physical overhaul: establishing bipedalism to distribute the upper body's center of mass directly along the vertical line of the spine. This is

mathematically proven by **[Formula 2: Biomechanical Energy Efficiency Formula]**:

$$\eta = \frac{B_r}{B_e \times (W_b \times \cos \theta)}$$

η : Biomechanical Efficiency

B_r : Effective Metabolic Energy Required by the Brain

B_e : Whole-Body Base Metabolic Rate

W_b : Weight of the Enlarged Brain

θ : Spine Angle relative to the Ground Vertical Line

(Note: $\theta < 90^\circ$ represents the limit state just prior to complete upright posture; simulations utilize values approaching 90°)

Mathematical Verification and Simulation of Formula 2 using Biomechanical Data

Plugging physical parameters from humans and chimpanzees into the formula clearly explains why humans had to adopt bipedalism to sustain an enlarged cognitive apparatus:

Non-Bipedal Ape Structure ($\theta = 15^\circ$, assuming enlarged brain $W_b = 1.4$ kg):

$$\text{Denominator Penalty Term} = 1.4 \times \cos(15^\circ) = 1.352$$

Interpretation: A bent spine subjects the back to direct horizontal load from brain weight, causing biomechanical efficiency (η) to plummet. This structure cannot sustain the energy leak, leading to extinction

Fully Bipedal Structure ($\theta = 89^\circ$, actual human brain $W_b = 1.4$ kg):

$$\text{Denominator Penalty Term} = 1.4 \times \cos(89^\circ) = 0.024$$

Interpretation: By trigonometric principles, as $\theta \rightarrow 89^\circ$, the brain weight penalty term approaches zero. Eliminating this mechanical overload enabled humans to stably power a massive brain ($B_r = 400$ kcal) without energy waste, securing a proportional energy conservation system.

3. Conclusion

This paper reinterprets the cognitive evolution of *Homo sapiens* not as a result of random environmental luck or innate physical superiority, but as an inevitable, paradoxical adaptation forced by extreme physical vulnerability and environmental danger.

Sequential analysis through mathematical modeling and ecological data reveals a clear domino chain of human evolution:

- [Genetic Weakness & Loss of Tree Habitats]
- [Savanna Isolation & Collective Imitation Network (Formula 1)]
- [Brain Expansion from Knowledge Spread]
- [Upright Bipedalism to Cancel Brain Weight Penalty (Formula 2)]

In conclusion, humans were the only species that satisfied the condition for high intelligence *because* they possessed biologically fragile bodies. While lions and chimpanzees lacked the pressing need to develop complex collective imitation due to independent physical defense, humans on the brink of extinction had to take the exclusive path of collective imitation and bipedalism. This prerequisite hardware foundation later enabled humans to utilize stored intelligence to invent tools and control fire, entering advanced civilization.

By mathematically demonstrating the interaction between physical deficit and biomechanical efficiency, this study offers a new paradigm regarding the origin of human cognition and civilization.

References & Data Sources

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